

Tanaya Sharma

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering | Minor in Artificial Intelligence (ML) Sept. 2024 – May 2028

- **Relevant Coursework:** Computer Organization, Digital Systems, Software Design, Deep Learning, Data Structures & Algorithms, Computer Graphics

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, SQL

Cloud & DevOps: AWS, Terraform, Docker, Infrastructure as Code (IaC), GitHub Actions, CI/CD, Git, Linux

Frameworks & Data: Spring Boot, Next.js, PyQt6/Qt, REST APIs, WebSockets, OAuth 2.0, PostgreSQL, Redis

Software: CMake, GoogleTest, Automated Testing, Debugging

Systems & ML: Embedded Linux, Multithreading, Firmware, Microcontrollers, PyTorch, OpenCV, SciPy

EXPERIENCE

Software Engineer

June 2026 – Present

Lodaria (YSpace) & PathwayR

Toronto, ON

- Architected and shipped a **full-stack evaluation platform** using **TypeScript, Python, Hugging Face, and REST APIs** to automatically assess free-form technical responses, including CAD and engineering submissions, across a **416-question bank**, reducing manual review workflows by **70%**.
- Designed and deployed **PostgreSQL infrastructure on AWS** supporting **10,000+ evaluation records**; implemented database indexing and query optimizations that reduced query latency by **35%**.
- Engineered real-time backend workflows using **WebSockets**, server-side state, and persistent database operations, improving production reliability through **95 automated tests and 18+ adversarial test cases**.

Founder & Engineer

May 2026 – Present

Sochy | Brampton Entrepreneur Centre

Brampton, ON

- Led end-to-end engineering across **full-stack software**, embedded C++, APIs, and hardware, delivering a connected E-Ink product and securing **\$3,000 in startup funding**.
- Built a full-stack web interface and backend for **device setup, management, and telemetry**, enabling users to connect devices while providing centralized access to data from connected devices.

Software Engineer

Sept. 2025 – Present

Blue Sky Solar Racing, University of Toronto

Toronto, ON

- Developed and debugged Python and **embedded Linux** systems for sensor integration and PCB bring-up, using oscilloscopes to identify hardware/software faults before vehicle integration.

Undergraduate Researcher

May 2025 – Oct. 2025

Computational Health and Interaction Lab, University of Toronto

Toronto, ON

- Developed C++ and MicroPython embedded software for a 13-channel control system, applying **multithreading** and microcontroller programming to synchronize hardware at **10 ms intervals**.
- Built a **Python, SciPy, and PyQt6** application for image processing, optimization, visualization, and hardware control, delivering an end-to-end research tool validated through hardware-in-the-loop testing.

Machine Learning Engineer

July 2024 – Sept. 2024

Omdena

Remote

- Built Python machine-learning and data pipelines from geospatial, weather, and satellite data for **8,200+ schools**, achieving **0.19°C MAE** and automated seven-day forecasting.

SELECTED PROJECTS

RiskRules | *Java, Spring Boot, Next.js, PostgreSQL, Docker, AWS*

- Built a full-stack Java platform using **Spring Boot, Next.js, and PostgreSQL**, demonstrating backend, database, and frontend proficiency through an explainable financial risk system.
- Deployed and automated infrastructure with **Docker, AWS, and Terraform**, demonstrating cloud deployment, CI/CD, and production systems experience.

Side B | *TypeScript, Next.js, PostgreSQL, Redis, Spotify API*

- Built and deployed a full-stack TypeScript application using Next.js, PostgreSQL, Redis, OAuth 2.0, and external APIs, validating production workflows with **105 automated tests**.